KAO LABS · MOBILE MANIPULATOR

Camera Recording Guide

How to record video from the robot's cameras, watch it back, and change the recording settings. Written for someone new to the project — no prior knowledge of the codebase needed.

`TidyBot++ reproduction • repo: chenchristian/tidybot2 • branch: christian_chen`

1 What this is

The robot has **three cameras**: one on the front of the base looking forward, and one on each arm's wrist. This guide covers recording color video from all three at once into dated folders, watching those recordings, and adjusting settings such as where files are saved and how fast they record.

CAMERA	WHERE	HARDWARE
Base	Front of the base, facing forward	Logitech C930e webcam
Left wrist	On the left arm near the gripper	Intel RealSense D405
Right wrist	On the right arm near the gripper	Intel RealSense D405

All three record **color video only**. The recording tools do not move the robot.

2 Before you start

You need two things from the team:

- The mini-PC's login (username and password).
- Your laptop connected to the same Wi-Fi as the robot (`NECL_5G`).

Then connect to the robot's onboard computer and open the project:

```
ssh eric@sixsevensupremacy.local
cd ~/tidybot2
conda activate tidybot2
```

Everything in this guide is run from that `~/tidybot2` folder with the `tidybot2` environment active. You'll know the environment is active when your prompt starts with `(tidybot2)`.

3 Record a video

```
python record_video.py
```

This opens all three cameras, waits for them to warm up, and starts recording. Press **Ctrl+C** when you want to stop. To record a fixed length instead and stop automatically:

```
python record_video.py --seconds 30
```

Each recording is saved to its own timestamped folder:

```
~/tidybot2/data/demos/20260902T134311/
    base_image.mp4
    left_wrist_image.mp4
    right_wrist_image.mp4
    data.pkl
```

You get one `.mp4` per camera. The `data.pkl` file holds timing information used later for training — you can ignore it for viewing.

> **Only one program can use the cameras at a time.** If the live monitor (Section 5) is running, stop it first or the recording will fail to open the cameras.

4 Find and watch your recordings

List recordings, newest first:

```
ls -t ~/tidybot2/data/demos
```

The mini-PC has no screen, so copy the videos to your own laptop to watch them. Run this **on your laptop** (not over SSH) — it grabs the most recent recording and opens the videos:

```
LATEST=$(ssh eric@sixsevensupremacy.local 'ls -t ~/tidybot2/data/demos | head -1')
scp -r eric@sixsevensupremacy.local:"~/tidybot2/data/demos/$LATEST" ~/Downloads/
open ~/Downloads/$LATEST/*.mp4
```

To fetch a specific recording, use its folder name instead of `$LATEST`.

5 Check the cameras before recording

The live monitor shows all three feeds in a browser so you can confirm framing and focus before you record. Start it on the mini-PC:

```
python camera_monitor.py
```

Then, in a browser on your laptop, open:

```
http://sixsevensupremacy.local:8000
```

One feed shows large; the others are thumbnails on the side — click a thumbnail to enlarge it. Each feed is labelled with its location and camera serial number. Press **Ctrl+C** on the mini-PC to stop the monitor when you're done, *before* you record.

> If the page won't load, make sure your laptop is on `NECL_5G`. If the name doesn't work, ask the team for the mini-PC's IP address and use `http://<that-ip>:8000` instead.

6 Change the settings

Every setting lives in one file: `record_video_config.py`. Open it in a text editor on the mini-PC:

```
nano ~/tidybot2/record_video_config.py
```

SETTING	WHAT IT CONTROLS
`DATA_DIR`	Where recordings are saved. Default `"data/demos"` (inside the project folder).
`CAMERAS`	Which cameras to record. Put a `#` at the start of a line to skip that camera.
`BASE_CAMERA_SERIAL` `LEFT_WRIST_CAMERA_SERIAL` `RIGHT_WRIST_CAMERA_SERIAL`	The serial numbers that identify each camera. Get current ones by running `python -m cameras`.
`RECORD_HZ`	How many frames per second are saved. Default `10`.
`LOGITECH_RESOLUTION` `REALSENSE_RESOLUTION`	Video size as `(width, height)`.
`REALSENSE_CAPTURE_FPS`	How fast the wrist cameras capture. Must be `30`, `60`, or `90`.
`LOGITECH_FOCUS`	Base camera focus. `0` = focus far away; use `100` if the fisheye lens is fitted.
`VIDEO_CODEC`	Video format. Leave as `"mp4v"` unless you know the mini-PC has an H.264 encoder.
`MAX_SECONDS`	Auto-stop after this many seconds. `None` means record until you press Ctrl+C.

Save the file (in `nano`: Ctrl+O, Enter, then Ctrl+X). The new settings take effect the next time you run `record_video.py`.

One-off overrides

To change something for a single recording without editing the file:

```
python record_video.py --seconds 30 --endpoint /mnt/nas/demos
```

FLAG	EFFECT
`--seconds N`	Stop after N seconds.
`--endpoint PATH`	Save this recording to PATH instead of the default.
`--record-hz HZ`	Use a different frames-per-second for this recording.

7 If something goes wrong

SYMPTOM	FIX
Recording fails to open a camera / `AssertionError`	The live monitor is probably still running. Stop it (Ctrl+C on the mini-PC), then try again.
“RealSense … not found”	Check the wrist camera’s USB cable (it needs a blue USB 3 port). Run `python -m cameras` to see what’s connected and its serial.
The video is all black	Lens cap still on, or the camera is pointed at nothing. Use the live monitor to check.
The monitor page won’t load	Confirm your laptop is on `NECL_5G`. Try the mini-PC’s IP address instead of its name.
“command not found: python”	The environment isn’t active. Run `conda activate tidybot2` first.

Quick reference

`ssh eric@sixsevensupremacy.local`	connect to the robot computer
`cd ~/tidybot2 && conda activate tidybot2`	open the project
`python record_video.py`	record (Ctrl+C to stop)
`python record_video.py --seconds 30`	record for 30 seconds
`ls -t ~/tidybot2/data/demos`	list recordings, newest first
`python camera_monitor.py`	start the live monitor → `http://sixsevensupremacy.local:8000`
`python -m cameras`	list connected cameras and serials
`nano ~/tidybot2/record_video_config.py`	change settings